

Getting Started

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To start using the features in the Custom Compiler environment as quickly as possible, browse the information and resources provided in this topic.

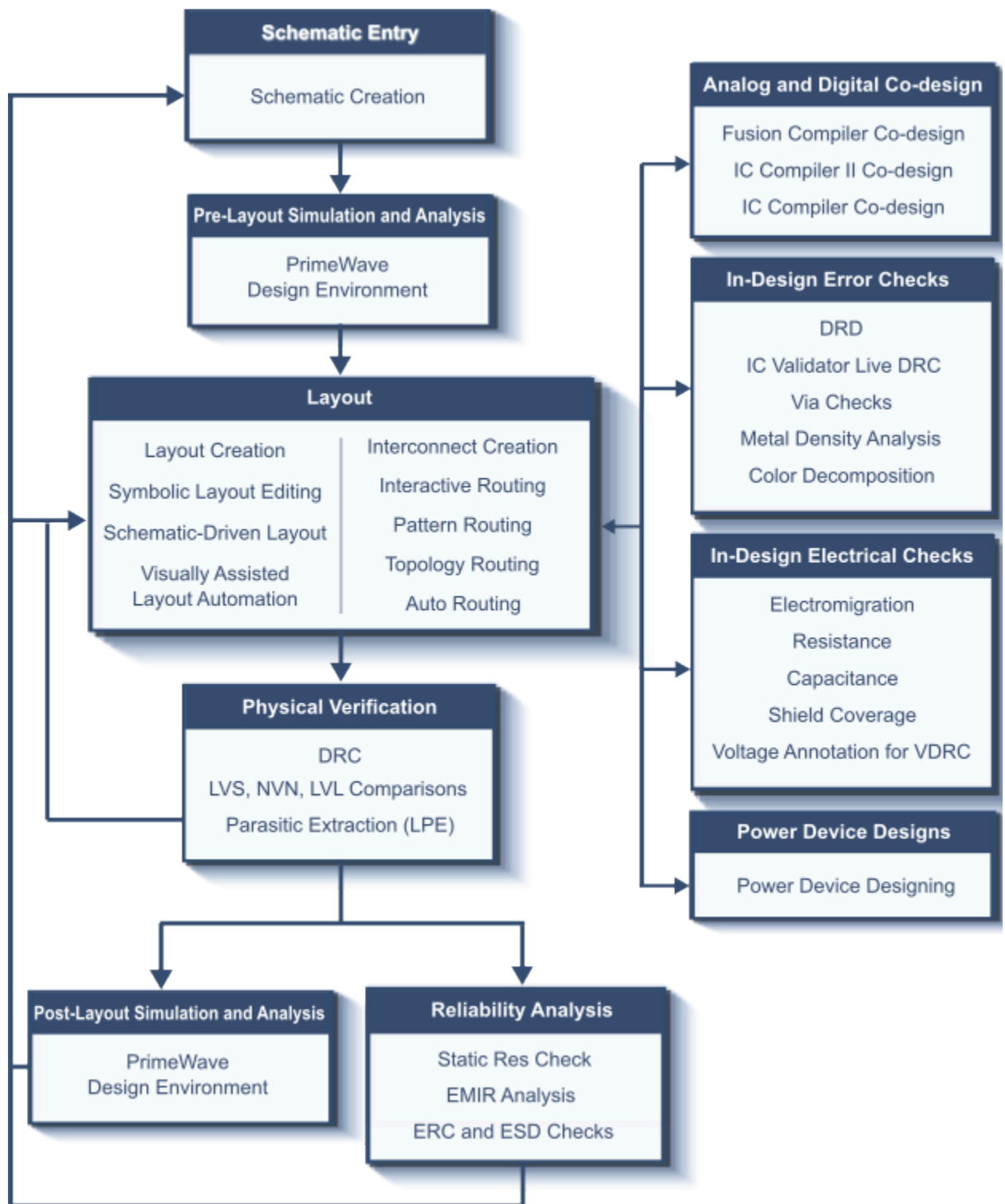
This topic covers the following:

- [Analog and Mixed-Signal IC Design Workflow](#)
- [Short Tutorial Videos](#)
- [Related User Documentation](#)

Analog and Mixed-Signal IC Design Workflow

Click a task title to view the topic.

For packaging designs, see Packaging Design Workflow.



Short Tutorial Videos

See the [Custom Compiler Short Tutorial Videos](#) article on SolvNetPlus for the latest list of available videos.

Note:

You must be logged in to SolvNetPlus for the link to connect directly to the article. If you are prompted to log in to SolvNetPlus upon clicking the link to the article, log in, then click the link again to reach the article.

To view the videos in context of design flow training, see the [Synopsys Learning Center Course Catalog](#).

Document	Description
Design entry	• Basic schematic editing • Add wire stubs to schematic designs • Add routes in the Schematic Editor • Auto chop wires • Create schematic blocks • Peek within schematic blocks • Create hierarchy configurations using the Hierarchy Editor • Compare designs • Edit multiple instances simultaneously in the Property Editor • Review designs • Search and replace design components • Create schematic overlays

Document	Description
Layout	• Initialize SDL • Create and use SDL presets • Create a block-level layout • Place devices with visually assisted automation • Create instance arrays using RPL commands • Create memory instances using RPL scripts • Use the Design Rule Driven (DRD) tool • Cross-select objects across assistants • Select objects under the cursor • Facilitate object selection with the Selection Table • Use the Clipboard assistant and toolbar • Display, hide, and manage LPPs using the Object/Layer Panel • Measure distance using rulers and dimension marks • Create wide alignment boundary • Align and distribute objects • Edit multiple instances simultaneously in the Property Editor (start at 0:51) • Stretch and split instance arrays • Create inverter PyCell using the UDD Assistant • Create customizable metal mesh PyCells using the UDD Assistant • Create and edit track region PCells • Create global track regions • Define tracks and track groups using the Track Region Assistant • Route device • Pattern routing • Create and extend bus routes • Create and manage design-view bookmarks • Retrace design operations with the Transaction History • Create layout templates and reuse the templates to manage design ECOs • Reuse templates and template libraries to create layouts • SDL ECO and auto-update between schematic and layout
In-design electrical checks	• Report Capacitance and Resistance (Parasitic Reporter) • Report Electromigration (Electrical Reporter) • Report Resistance (Electrical Reporter) • Report Capacitance (Electrical Reporter)
Reliability analysis (using the SAE tool*)	• Set up and run reliability analysis simulations • Use the Reliability Analysis Results Browser • Run what-if analyses

Document	Description
Simulation, analysis, and debug (using the SAE tool*)	• Set up multiple testbenches • Set specifications and use the SAE Calculator • Run corner simulations • Analyze corner simulation results • Noise analysis and noise summary reports in SAE • Operating Point (OP) Analysis and OP Reports in SAE • Sequential simulation in SAE • Run parametric simulation • Use the Circuit Check graphical user interface • Use the Estimated Parasitics Assistant • Compare results in SAE • Use charts in SAE • View probability distribution waveforms with SDE noise analysis *The PrimeWave Design Environment is now the default simulation environment in the Custom Compiler tool. To use the legacy simulation environment (SAE), set the environment variable <code>SYNOPSYS_CUSTOM_LEGACY_SIMENV=1</code>.
Analog and digital co-design with IC Compiler II and Fusion Compiler**	• Add IC Compiler II library in the Custom Compiler environment • Basic preroute editing • Implement digital block • Create frame view (basic instructions) • Create frame view for an analog block in an IC Compiler II top-level design • Region-based editing • Cloning • Report and check resistance • Report and check capacitance • Top-level ECO editing ** The videos are shown using the IC Compiler II tool and designs, but the instructions are also applicable to Custom Compiler and Fusion Compiler co-design flows.
Physical verification and extraction	• Analyze design density • Extract parasitics (GPD-based queries) • Run DRC with the integrated IC Validator tool • Run layout-versus-schematic comparison using IC Validator in the Custom Compiler environment

Related User Documentation

Design Phase	User Documentation
Co-designing	3DIC Compiler Fusion Compiler IC Compiler II IC Compiler
Packaging designs	3DIC Compiler
Photonics	OptoCompiler
Physical verification and extraction	StarRC IC Validator
Simulation	PrimeSim Continuum PrimeSim Pro and PrimeSim SPICE PrimeSim HSPICE PrimeSim XA VCS Verdi
